ModelIt!

Student Activity Pack

How Do Pandemics Spread?

Modeling Epidemiological Dynamics, Transmission Networks, and Intervention Strategies

Grade 12L3 | NGSS HS-LS2-8 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What does it mean when scientists say a virus has an R0 of 3, and why does that matter for how fast a pandemic spreads?**

2. **Why do you think some pandemics kill thousands while others kill millions, even when caused by similar types of viruses?**

3. **What interventions did your community use during the COVID-19 pandemic, and which ones do you think were most effective?**

4. **What do you think 'herd immunity' means and how is it related to vaccination?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Viral Transmissibility (R0)
- Population Density
- Initial Immune Fraction
- Contact Rate
- Infection Detection Rate
- Quarantine Compliance
- Vaccination Coverage Rate
- Healthcare System Capacity
- Mutation Rate
- Cumulative Mortality

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Uncontrolled Spread

R0: 3 | Population Density: High | Initial Immunity: 0% | No interventions

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Early vs. Late Intervention

R0: 3 | Interventions start at Week 2 vs. Week 6

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Variant Emergence

R0: 3 initially | Mutation Rate: High | Immune evasion variant appears at Week 20

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Economic Impact Index — The cumulative economic damage from both the pandemic itself and the interventions used to control it — business closures, unemployment, supply chain disruption, and healthcare costs that disproportionately affect low-income communities and create pressure to relax interventions prematurely

Misinformation Spread Rate — The velocity at which false health information propagates through social media and community networks — misinformation reduces vaccination uptake, quarantine compliance, and trust in public health authorities, effectively increasing the virus's transmission advantage

Genomic Surveillance Capacity — The ability to sequence viral genomes from patient samples to detect emerging variants early — countries with strong genomic surveillance can identify immune-evasive or more transmissible variants weeks before they become dominant, enabling proactive rather than reactive public health responses

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When a new virus emerges, it can circle the globe in weeks, infecting millions before scientists even identify it. How do mathematical models predict pandemic trajectories, and why do small differences in transmission rate or intervention timing determine whether thousands or millions die? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why exponential growth makes early intervention timing far more important than intervention intensity?**

2. **What does the model reveal about why healthcare capacity is a mortality multiplier rather than just a treatment variable?**

3. **Why is herd immunity a moving target rather than a fixed threshold, and what does this mean for vaccination strategy?**

4. **How would adding economic impact or misinformation to your model change the predicted effectiveness of interventions?**

5. **What are the limitations of modeling pandemic spread when individual human behavior varies enormously and changes over time?**

8 STEM Challenge: Design a Pandemic Preparedness Plan

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YOUR ENGINEERING MISSION

Design a comprehensive pandemic preparedness plan for a city of 1 million people that minimizes cumulative mortality while balancing economic, social, and equity considerations, using model data to justify intervention timing and intensity.

THE SCENARIO

A novel respiratory virus with an estimated R0 of 4 has been detected in three countries. Your city of 1 million people has zero confirmed cases but the virus is expected to arrive within 2-4 weeks. No vaccine exists yet and development will take 12-18 months. Your team must design a preparedness plan that specifies exactly when and how to implement testing, quarantine, social distancing, healthcare surge capacity, and eventually vaccination — and justify every decision with model data.

Engineering Design Thinking

1. WHAT IS YOUR INTERVENTION TIMELINE AND WHAT MODEL EVIDENCE SUPPORTS EACH TRIGGER POINT?

2. HOW DOES YOUR PLAN ADDRESS EQUITY — ENSURING THAT QUARANTINE COMPLIANCE IS FEASIBLE FOR LOW-INCOME WORKERS AND THAT VACCINATION REACHES UNDERSERVED COMMUNITIES?

3. WHAT ECONOMIC SUPPORT MECHANISMS DOES YOUR PLAN INCLUDE TO MAKE SOCIAL DISTANCING FINANCIALLY SUSTAINABLE?

4. HOW WOULD YOU ADAPT YOUR PLAN IF A MORE TRANSMISSIBLE VARIANT EMERGED MIDWAY THROUGH IMPLEMENTATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists who track and model disease outbreaks earn $60,000-$120,000/year at agencies like the CDC, WHO, and state health departments. Biostatisticians who build mathematical models of pandemic spread earn $75,000-$140,000/year. Public health emergency preparedness directors coordinate city and county pandemic response, earning $80,000-$150,000/year. Infectious disease physicians who treat patients during outbreaks earn $200,000-$350,000/year.